Instruction

Criteria for evaluating the difference between two mathematical solutions include:
1. **Different methods**: If the methods used to arrive at the solutions are fundamentally different (e.g., algebraic manipulation versus geometric reasoning), the solutions can be considered distinct.
2. **Different intermediate steps**: Even if the final results are the same, the solutions can be considered different if the intermediate steps or processes vary significantly.
3. **Different assumptions or conditions**: If two solutions rely on different assumptions or conditions, they are likely to be distinct.
4. **Different generalizability**: A solution might generalize to a broader class of problems, while another might be specific to certain conditions. In such cases, they are considered distinct.
5. **Different complexity**: If one solution is significantly simpler or more complex than the other, they can be regarded as essentially different —even if they lead to the same result.
6. **Too simple questions**: If a problem is too simple, e.g. "How many positive factors of 36 are also multiples of 4?", it is hard to be novel (**mark as not novel**).

Problem

Let $ABCD$ be a cyclic quadrilateral. Prove that there exists a point X on segment $\overline{BD}$ such that $\angle BAC = \angle XAD$ and $\angle BCA = \angle XCD$ if and only if there exists a point Y on segment $\overline{AC}$ such that $\angle CBD = \angle YBA$ and $\angle CDB = \angle YDA$.

problem image (ignore if empty)

Reference Solutions

Solution 1. Note that lines AC, AX are isogonal in $\triangle ABD$, so an inversion centered at A with power $r^2 = AB \cdot AD$ composed with a reflection about the angle bisector of $\angle DAB$ swaps the pairs (D, B) and (C, X). Thus, $\frac{AD}{XD} \cdot \frac{XD}{CD} = \frac{AC}{BC} \cdot \frac{AB}{CA} \implies (A, C; B, D) = -1$ so that $ACBD$ is a harmonic quadrilateral. By symmetry, if Y exists, then $(B, D; A, C) = -1$. We have shown the two conditions are equivalent, whence both directions follow.

==========
Solution 2. All angles are directed. Note that lines AC, AX are isogonal in $\triangle ABD$ and CD, CE are isogonal in $\triangle CDB$. From the law of sines it follows that $\frac{DX}{XB} \cdot \frac{DE}{ED} = \left(\frac{AD}{DB}\right)^2 = \left(\frac{DC}{BC}\right)^2$. Therefore, the ratio equals $\frac{AD \cdot DC}{DB \cdot BC}$. Now let Y be a point of AC such that $\angle ABE = \angle CBY$. We apply the above identities for Y to get that $\frac{CY}{YA} \cdot \frac{CE}{EA} = \left(\frac{CD}{DA}\right)^2$. So $\angle CDY = \angle EDA$, the converse follows since all our steps are reversible. Beware that directed angles, or angles $\bmod 180$, are not standard olympiad material. If you use them, provide a definition.

Proposed Solution

Forward Direction (⇒): 1. **Assume Existence of X:** Let X be on BD such that $\angle BAC = \angle XAD$ and $\angle BCA = \angle XCD$. 2. **Cyclic Quadrilateral Properties:** Since $ABCD$ is cyclic: - $\angle BAC = \angle BDC$ (subtends arc BC). - $\angle BCA = \angle BDA$ (subtends arc BA). Thus, $\angle XAD = \angle BDC$ and $\angle XCD = \angle BDA$. 3. **Establish Similar Triangles:** - In $\triangle XAD$ and $\triangle BDC$: $\angle XAD = \angle BDC$ and $\angle AXD = \angle BCD$ (since $\angle BCD = \angle BAD$, cyclic). Hence, $\triangle XAD \sim \triangle BDC$ (AA similarity). - In $\triangle XCD$ and $\triangle BDA$: $\angle XCD = \angle BDA$ and $\angle CXD = \angle BAD$. Hence, $\triangle XCD \sim \triangle BDA$ (AA similarity). 4. **Proportional Relationships:** From similarity: - $\frac{XA}{BD} = \frac{AD}{BC} \implies XA = \frac{AD \cdot BD}{BC}$. - $\frac{XC}{BD} = \frac{CD}{BA} \implies XC = \frac{CD \cdot BD}{BA}$. 5. **Menelaus' Theorem on $\triangle ABD$:** With transversal XC: $\frac{BX}{XD} \cdot \frac{DC}{CA} \cdot \frac{AY}{YB} = 1$. Substituting the proportional relations from step 4 and solving for $\frac{AY}{YB}$, we find Y on AC such that $\angle YBA = \angle CBD$ and $\angle YDA = \angle CDB$. **Reverse Direction (⇐):** Symmetrically, assuming Y exists on AC, analogous steps with Menelaus' theorem and similarity establish the existence of X on BD. **Conclusion:** The equivalence follows by the symmetric application of similar triangles and Menelaus' theorem, ensuring mutual existence of X and Y under the given angle conditions.

Correctness

The solution is correct.

○ correct ○ incorrect

Novelty

The solution is novel compared to reference solutions.

○ novel ○ not novel

Feedbacks/Questions

If any part of this HIT is confusing or if you have any feedbacks or question for us, please let us know below.

Figure 8: Annotation interface for CreativeMath.

E Performance Summaries

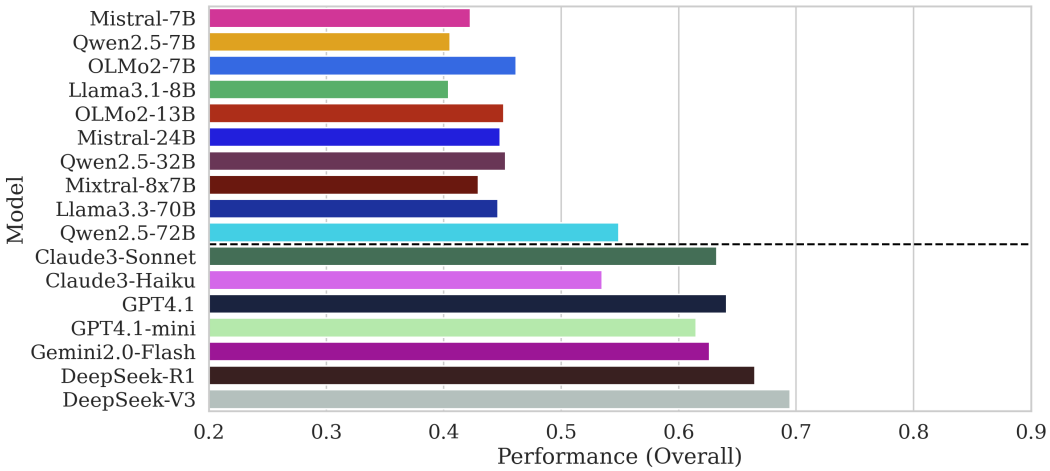

Figure 9: Overall performances.

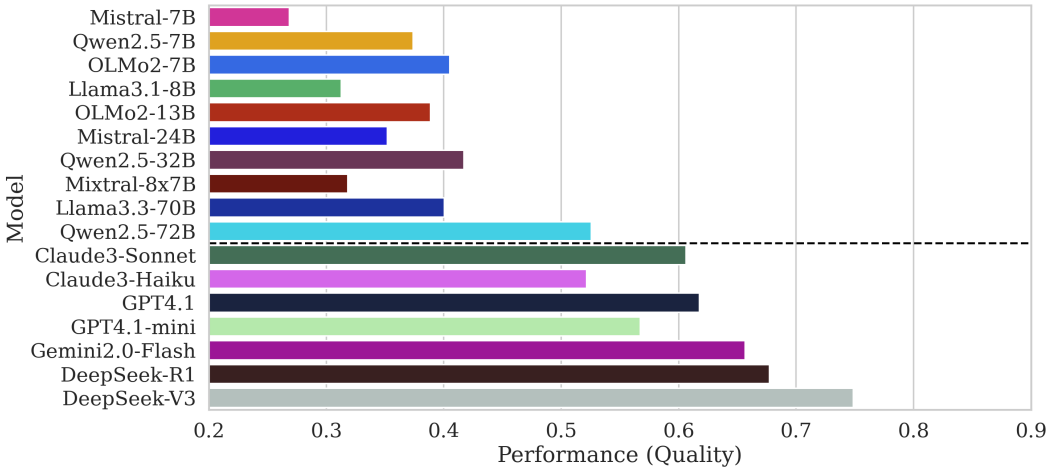

Figure 10: Performance on quality dimension

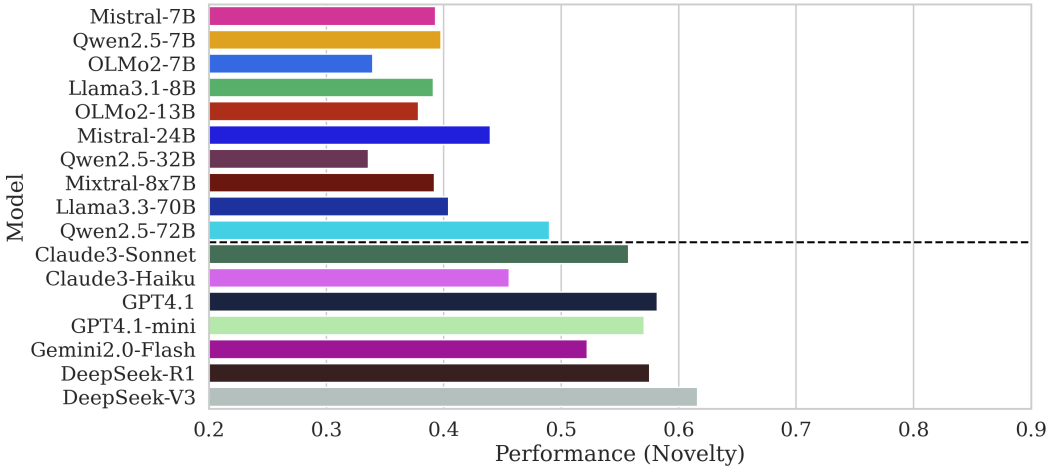

Figure 11: Performance on novelty dimension

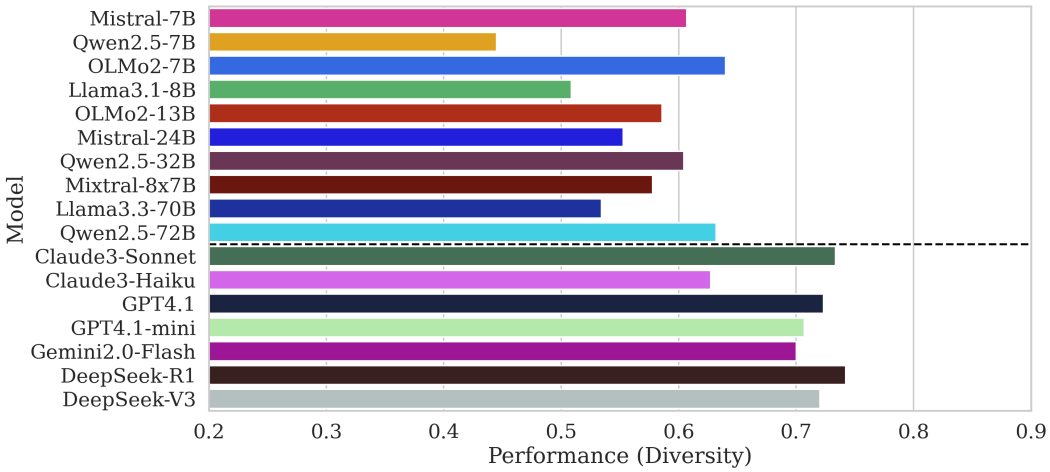

Figure 12: Performance on diversity dimension